Verification round 2: PCB DFM, fab package and mechanical integration

Reviewer: independent DFM / fab-package and adversarial mechanical-integration review. Date 2026-11-02. Repo `/home/user/reading_light`, branch `claude/great-shannon-bikrur` @ `15d1f96` (clean). Nothing in the repo was edited. Scratch work (scripts, renders, DRC re-runs): `/tmp/claude-0/-home-user/7ba8e87c-d4ee-5a66-9891-51bde4e0a9a7/scratchpad/review2/`.

Method. Gerber/drill files were read directly (apertures, G85 slots, outlines). The two `.kicad_pcb` files were loaded read-only with `pcbnew`, which gave pad geometry, nets, zone fills, radial extents, mask webs and via-in-pad. DRC was re-run on a scratch copy of the LED board with 0.2 mm clearance. Layers were rasterised with `kicad-cli pcb export pdf`, and the assembly PDFs and renders were inspected. Datasheet land patterns were rendered from the PDFs in `scratchpad/ds/`. Print STLs were parsed with numpy to get vertex extents and connected components. Packlib and moonlib values came from the `cad/parts` sources.

Result: 3 BLOCKER, 3 MAJOR, 12 MINOR, 7 NOTE. The gerbers themselves would be accepted by PCBWay. What fails:

- The main board's battery-lead slot cannot pass the battery's plug (a PCB change, so do it before ordering).
- The printed back plate cannot be fitted into the front shell.
- The gooseneck thread end is not controlled against the LED board.
- The RGB LED polarity instructions are wrong, so Matt could fit both RGB LEDs rotated.

Findings

ID	Sev	Finding	Evidence	Recommended fix
R2-D1	**BLOCKER**	**The battery lead cannot reach J2 through the slot.** The lead-slot is a closed internal cut-out of 3.0 × 2.8 mm (r 0.6). The pouch sits behind the PCB and its PHR-2 plug must mate with J2 on the PCB top, so either the plug or the pouch has to pass through the slot. The BAT0004 lead is pre-terminated, and the PHR-2 housing is 5.8 × 4.5 mm in section, so neither fits. The only ways through are cutting the lead or de-pinning the SPH contacts (1.5 × 2.08 mm) from the housing with the cell live. That risks shorting the cell and re-inserting the contacts reversed (Q3 does not block a reversed cell while USB is present, round1 m2). Handoff 05 §4.2 "Plug J2, with the lead running through the slot" cannot be done as written	`hardware/main-board/design.py` `SLOTS=[(17.3,52.2,3.0,2.8,0.6)]`. Edge_Cuts gerber: closed loop x 15.8–18.8, y 50.8–53.6. JST PH catalogue p.2: PHR-2 B = 5.8, depth 4.5, length 6.35 + 0.5. SPH-002T: 1.5 × 2.08. `bom/BOM.csv:30`: BAT0004 "JST-PH 2.0 lead". research §B: 100 mm lead	Fix before ordering. the opening pass th ≥ 6.4 × 5.0 mm. On that works is x 10.0- y 49.6–54.6, inside t plug keep-out (wher plug lies anyway). T keeps 0.3 mm to the pad (x ≤ 9.65) and t J1 B12 pad (y ≥ 54.9 and clears D1 (y ≤ 4 Move the four GND s vias there and re-ru Alternatively, open t to a board edge. If t board is ordered as document the de-pi procedure in handof (tape one contact w handling the other; polarity again, E1), c the lead and crimp a PHR-2

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R2-D2	**BLOCKER**	**The `pack_back` plate cannot be inserted into `pack_front`.** The plate fits inside the shell walls: plate ±17.65, shell inner wall ±17.85. Its battery-corner locators, however, are placed from `BAT` width 36.0 rather than from the cavity. They reach |X| = 18.9, which is 1.05 mm into each side wall, at four places (Y −30.85…−24.85 and 27.65…33.65, Z 1.8–3.8). The bottom-end locators reach Y −31.75, 0.4 mm into the bottom wall (inner face −31.35). The bay itself (36.0) is also wider than the cavity (35.7), so a 35.5 mm pouch has 0.1 mm per side. No check covers this: the pack_back context holds only the battery, and pack_front holds only the PCB	`print/pack_back.stl`: bbox X ±18.9, vertices (−18.9, −30.85…33.65, 1.8–3.8) and (±18.4, −31.75). `print/pack_front.stl`: back-rim inner |x| = 17.85, inner bottom y = −31.35. `cad/parts/pack_back.py` lines `x_out = sx*(bw/2+0.8)` and `yy - sy*0.4`. `packlib.IN_X = 17.85`, `BAT = (36.0, …)`	Derive the locators f the cavity: keep |X| − clear, and keep th bottom locators insi IN_Y_BOT + clear. Th walls already locate mm pouch, so the lo side locators can sin be dropped. Either s `BAT` W ≤ 35.5, or w the shell (CLEAR/IN_ the cell measures w (VERIFY M7). Add a `clear_of` check of pack_back against pack_front in the pip Re-export `print/`
R2-D3	**BLOCKER** (conditional; thread length UNVERIFIED)	**Gooseneck thread end vs the LED board.** Nothing limits how far the M10 thread protrudes into `moon_base`. The stack is: floor Z 0–2, nut 2–5, board bottom at 7.2. U1 (TLC59116, TSSOP-28, 1.2 mm) hangs from the board centre, directly over the tube bore, with its underside at Z 6.0. LampSpares M10 goosenecks have 9 mm threads per end (268 − 250 mm in `docs/research`), and a 9 mm thread puts the tube end at Z 9.0, above the board. The board cannot seat, and the bayonet would press the tube's 3.4–5.0 mm-radius ring onto U1's output pins, shorting them to each other and to the metal neck. Even with the thread flush with the nut top, the four wires leave the tube axially under U1 and must turn 90° within 1.0 mm (Z 5.0 → 6.0). That is not possible with heat-shrink, and marginal with bare 28 AWG. The cable notch (Z 2–5, cut into the printed ring) cannot be reached, because the wires are inside the metal nut's bore. So handoff 05 §3.3 "Wires out through the cable notch" is not physically possible. VERIFY M1 measures only the thread diameter and the nut	`cad/parts/moon_base.py dims()`: floor_top 2.0, nut_top 5.0, ledge_z 7.2, `UNDER_BOARD_DEPTH` 2.2. LED board U1 at (50, 50) B, body 9.7 × 4.4 + leads to ±3.2 mm. TI PW package max height 1.2 mm. `docs/research/neck-battery-pcb-pricing.md:54` (250 mm excl. / 268 mm incl. threads). Round1 adversarial m-6 suggested "thread end flush with or below the nut", but this was never added to VERIFY or the handoff	A CAD-only fix, with board respin. (1) Ra `UNDER_BOARD_DEPTH` 4.5 mm. Rim, ring a moon move up with board, so the optics is unchanged and th gets ~2.3 mm taller Add VERIFY "thread protrusion above the ≤ FLOOR_T + NUT_T mm; cut or file the t to length". (3) Add a printed PETG top-ha insulating disc over nut (round1 m-6). (4 Move the cable notc above the nut top so can leave the bore a to J1 (north) at Z 5–6 Fix the handoff word

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R2-D4	**MAJOR**	**The Würth RGB (D9, D10) polarity instructions are wrong.** Handoff 05 §2.2 says "Würth RGB (pin 1 = common anode, corner mark)". On the 150141M173100, the **polarity mark (chamfer/dot) is at pin 3, the red cathode**. Pin 1 (anode) is the adjacent corner on the same short side. The board silk carries both a chamfer at **pad 3** (matching the package) and a "1" at **pad 1**, so following the text puts the mark on pad 1. Both RGB LEDs would then sit rotated 90°: VLED lands on the red cathode and the anode lands on K_B, so both are dark or wrong (sink driver; probably not destructive). The PCBWay BOM note on D9/D10 also says "pad 1 = cathode (large pad on 2835)", which is wrong: pad 1 is the anode. An assembler following it would mis-rotate them	Würth datasheet p.1: top view "Polarity Mark (Cathode; red LED)" at pin 3; circuit "Common Anode + 1, 2 B−, 3 R−, 4 G−". KiCad `LED_RGB_Wuerth-PLCC4…` silk chamfer at (−2.5, −1.2), next to pad 3 (−1.55, −0.7), and text "1" at (3, −0.7), next to pad 1. `fab/led-board/v0.1/renders/led-board_assembly_top.png` shows both marks. `bom_pcbway.csv` line 3	Handoff: "Würth RGI the package's chamfered/dot corne 3, red cathode) on t chamfer. Pin 1, the common anode, goe the pad marked '1', is VLED." Fix the BO for D9/D10 to "pad 1 common anode; pac polarity mark = pin Optionally add a "+' to pad 1 on F.Silk. Th (0603) note also me "2835", see R2-D9
R2-D5	**MAJOR**	**J2 header height vs the front face.** The S2B-PH-SM4-TB is **5.5 mm** tall per JST. `TOP_CLEAR` is 5.6 mm, which leaves 0.1 mm of nominal margin before solder paste (~0.05–0.1 mm), boss-height and print tolerance. The bosses set the PCB-top-to-face distance, so a header touching the face either stops the screws seating or bends the PCB and loads J2's joints. VERIFY M8 and `pack_front.VERIFY["jst_ph_height"]` assume ≤ 5.2 mm, and packlib comments "~5 mm VERIFY". The datasheet already settles it at 5.5. The fab STEP has no J2 model (R2-D17), which is why the 3D check did not catch it	JST PH catalogue p.3, "SMT type Header / Side entry": 5.5 overall height. `cad/parts/_lib/packlib.py`: `TOP_CLEAR = 5.6`. J2 at board (5, 53), header body x ≈ 1.7–9.3, y 50–56	Cut a 0.8 mm-deep into the inside of the face over J2 and its board x 0–16, y 48–5 pack X −17.5…−1.5 −27…−17. The 1.8 face keeps 1.0 mm. Alternatively raise TOP_CLEAR to ≥ 6.2 re-derive the button Update VERIFY M8 t
R2-D6	**MAJOR**	**The only on/off control is hard to reach.** The PCM12 actuator tip is 1.0 mm past the board edge, at board x 36.0. With CLEAR 0.35 and a 1.8 mm wall, the tip sits **1.15 mm below the outer surface** in a plain 2.6 × 5.0 mm rectangular window. A fingertip cannot reach it, so it needs a fingernail or a pen tip. `pack_front` B8 promises "a 45 deg outer chamfer (a pyramid frustum cut)", but the code only extrudes the rectangle, and the STL confirms there is no chamfer. The B8 claim checks only that the window is open	`pack_front.stl`: window vertices only at |x| 17.85 and 19.65, y 0.5–5.5, z 5.2–7.8. No chamfer. PCM12 datasheet: actuator 1.3 wide, protrudes 1.5 beyond the body, travel 1.5. KiCad footprint: body front at local y +1.6, so board x 34.45, and actuator tip at local +3.15, so board x 36.0	Implement B8 as wr a 45° countersink fr outer face, e.g. 5.0 at the inner face ope to 7.4 × 5.0 at the o face. Thin the wall lo to ~0.8 mm around window, as was don USB-C. Add a claim t the knob tip is ≤ 0.5 below the local oute surface. A printed kn extender is an alter
R2-D7	MINOR	`fab/main-board/v0.1/FAB_NOTES.md` says "**Plated slots: none**", but the PTH drill file contains **4 plated oval slots**, 0.6 × 1.2 mm (G85), for the USB-C shell pegs. Width 0.6 is above PCBWay's usual 0.5 mm plated-slot minimum (UNVERIFIED against today's capability page), but a CAM query is likely if the notes contradict the files	`main-board-PTH.drl`: T2 C0.600, X13.18Y-55.3G85X13.18Y-55.9 ×4	FAB_NOTES: "Plated 4 × 0.6 × 1.2 mm (U shell), in the PTH dri

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R2-D8	MINOR	LED board FAB_NOTES (and the design RULES/netclass) state "min via 0.5 / 0.25 mm". The board actually has **29 vias at 0.6 / 0.30 mm**, and there is no 0.25 mm hole. Entering "0.25 mm min hole" on PCBWay's form can move the order into a smaller-drill (more expensive) class for no reason	`led-board-PTH.drl`: only T1 C0.300. pcbnew: vias {(0.6, 0.3): 29}. `led-board.kicad_pro` netclass via 0.5/0.25	FAB_NOTES: "min vi 0.3 mm; min hole 0. min space 0.18 mm (measured 0.186)". the netclass with wh routed
R2-D9	MINOR	CPL/BOM hygiene. (a) LED-board CPL lists **J1** (cable solder pads, not a component, `in_bom=False`) at Bottom 180°, but J1 is absent from the BOM, which will trigger an assembly-quote query. (b) The D2 note says "polarity: pad 1 = cathode (large pad on 2835)" on an 0603. The cathode part is right; the reference is wrong. (c) J1 USB-C is typed "SMD", but it has four plated shell slots needing THT solder. (d) The JLCPCB BOMs have an empty LCSC column (known; handoff 01 says fill it at quote time). DNP J4, TP1–6 and H1–2 are correctly excluded. Bottom-side LED-board parts are correctly flagged "Bottom". All MPNs match `hardware/parts.py`	`fab/led-board/v0.1/cpl_pcbway.csv` row J1. `bom_pcbway.csv` main line 9, LED line 3. pcbnew: J1 `excl_pos=False`	Set `exclude_from_pos_` on the LED J1 footpr Make the polarity nc per part type (2835: pad = cathode"; Wü see R2-D4; 0603: "c mark to pad 1"). Ma as "SMD + THT"
R2-D10	MINOR	The assembly drawings and LED-board silk are weak for hand placement. `assembly_top/bottom.pdf` are 1:1 on A4: the 26 mm board is ~2.6 cm on the sheet. The title block is empty (no title, rev or date). They carry reference designators only, with **no values**, so the 2700 K positions (D1/D3/D5/D7) and 5000 K positions (D2/D4/D6/D8) are not shown, and handoff 05 §2.2 tells Matt to find them there. DNP J4 is drawn as a normal part, not crossed out. The F.Silk references "D6" and "D7" sit directly under "D9"/"D10" near the centre, nearer the RGBs than their own LEDs. A swapped 2700/5000 K LED is silent: firmware drives each output as WW/CW from `board_map.h`, so colours come out wrong	Rendered `scratchpad/review2/img/asm_led-board_top-1.png`, `asmz_led_top-1.png`, `svg/led_top-1.png`	Re-export the assem drawings at 3:1 or fi page, with values (`%` and `--crossout-DN` `footprints-on-fab` `layers`, and fill the block. Add "W"/"C" ("27"/"50") silk mark beside each 2835 ar move the D6/D7 references next to tl LEDs. An interactive BOM would help bot boards
R2-D11	MINOR	The DOWN button cap (SW4) can rub the H2 mounting boss. The cap flange is Ø9.6 (r 4.8) and the boss is Ø5.0 (r 2.5), with centres 7.5 mm apart: **0.2 mm nominal gap**. The cap has ±0.2 mm lateral play (Ø7.0 in a Ø7.4 hole), and both are printed PETG overlapping in Z (flange at PCB top + 4.3…5.5 mm, boss full height). A dragging or sticking DOWN button is likely. SW2 vs H1 has 0.76 mm	`button_caps.py` `Cylinder(4.8, FLANGE_T)`. `packlib.BOSS_D = 5.0`. SW4 at board (17.5, 39), H2 at (17.5, 46.5)	Use a flat on the cap flange facing H2, or reduce the flange to (r 4.5 gives 0.5 mm) move H2 1 mm dow 47.5; it still clears th slot). Add a clear_of between caps and b

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R2-D12	MINOR	Hand assembly of the double-sided LED board. (a) Nothing supports the board for the second-side stencil print or reflow. With U1, U2 and C2 on the bottom (up to 1.45 mm), the 26 mm board will not lie flat under a frameless stencil. (b) The 2835s have bottom-only terminations (KiCad pads extend 0.25 mm past the body), so the handoff's "hot air **or the iron**" really means hot air or a plate. (c) "The bottom-side parts must not reflow again" cannot be achieved on a 1.0 mm, fully poured board under hot air. It is also unnecessary, because surface tension holds the TSSOPs. (d) Stencil thickness is not specified for 0.65 mm-pitch TSSOP + 2835. The bottom-first order is correct	Handoff 05 §2. FAB_NOTES LED "Stencils: top AND bottom (frameless)". KiCad LED_PLCC_2835 pads ±2.0 vs body ±1.75	Print a simple PETG 26.3 mm pocket 1.0 with a Ø23.5 × 2 mr relief for the bottom which supports the l only on the part-free 13 band. Use it for t stencil and for a sec hot-plate pass. Orde mm stencils, with to bottom on one foil if cheaper. Reword har §2.2 to "hot air or hc plate on the jig; the bottom parts will re- but stay put"
R2-D13	MINOR	The PCM12SMTR is rated **255 °C max** solder temperature. A hobby hot-plate SAC305 reflow often overshoots to 255–265 °C, and handoff 05 §1.3 allows the mechanical parts "through the same reflow"	PCM12 datasheet p.1 "SOLDERABILITY: 255C max."	Handoff: fit SW1 by after reflow (or use l paste, peak ≤ 230 °
R2-D14	MINOR	USB-C plug clearance. The receptacle face is flush with the board edge, so it sits **1.55 mm behind the outer surface** (0.35 gap + 1.2 mm thinned wall). The plug overmold must enter the 12.8 × 7.0 window, which has 1.5 mm 45° corner chamfers. An overmold at the USB-C maximum (12.35 × 6.5) passes those chamfers only if its corner radius is ≥ 1.75 mm. Square-cornered cables will stop up to ~1 mm short of full mating. Radial margin is 0.2–0.25 mm before PCB position and print tolerance	KiCad USB4125 footprint: Dwgs "PCB Edge" at local y 3.4 = body front, so board y 62.0 = edge. `pack_front.py` `_chamfered_rect(12.8, 7.0, 1.5)`, `WALL−1.2` pocket	Open the window to × 7.6 with ≤ 0.8 mn chamfers, or a stadi shape. Check the ca Matt will actually us to VERIFY M8)
R2-D15	MINOR	The LED board can rattle in the moon base. The pocket depth is BOARD_T + 0.01 = 1.01 mm, and the ring rests on both the board band and the cup step at Z 8.21. A board at the low end of the 1.0 ± 0.1 mm tolerance (0.9 mm) leaves the ring sitting on the step and the board loose with 0.11 mm play. Bayonet preload cannot reach it. VERIFY `ring_preload` exists but treats it as a shim job	`moon_base.py`: pocket `Cylinder(POCKET_R, BOARD_T + 0.01)` from ledge_z. `moon_ring.py`: OD 33.2 bears on the step r 13.2–16.6	Make the pocket dep 0.85–0.9 mm (board proud of the step), s ring always bears or board band and the bayonet's 0.35 mm drop gives preload. Measure the deliver board (VERIFY)
R2-D16	MINOR	Handoff 05 §4.3 says "Anchor the cable to the PCB edge ... about 10 mm before the connector". There is no such length of cable on the PCB. The J3 plug's rear sits ~2 mm past the board's top edge (pack Y ≈ 33), and the gooseneck bore opens at Y 35, so the cable is never alongside the PCB	`pack_front.py`: `_teardrop_y(..., K.PCB_H/2 + 4.0, ...)` (bore from Y 35). J3 header face at board y ≈ 0.6–0.95	Replace it with: "lea 10–15 mm service lo the neck region and cable with a dab of silicone at the bore entrance". Optionall a printed cable clip neck block

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R2-D17	MINOR	The fab STEP files are incomplete. `main-board.step` has **no model for J1 (USB4125), J2 (PH header) or F1**, and `led-board.step` has **no 2835 LEDs**: the cloud KiCad install has no `Connector_USB/JST/Fuse/LED_SMD` 3D libraries. Handoff 01 §1.4's 3D check, and any enclosure check done from the STEP, will miss the tallest part (J2, R2-D5)	`grep PRODUCT(` on both STEP files: 13 and 7 part types, none of these	Re-export the STEP Matt's machine with 9's 3D libraries befo using it for fit check say so in FAB_NOTE
R2-D18	MINOR (UNVERIFIED)	The battery lead path is not modelled. The bay leaves 0.5 mm (bottom) and ~1 mm (pouch 64.0 in a 64.5 bay) at the ends, and the pouch's PCM end must fold its 100 mm lead back over the pouch, into the 1.6 mm gap shared with Kapton and the two M2 heads (1.3 mm). The slot is 9.6 mm from the bottom bay end (pack Y −21.2), and the excess lead (~40–80 mm) has to be stowed somewhere	`packlib`: Z_BAT_TOP 8.0, Z_PCB_BOT 9.6. `pack_back.py`: `bay_y0 = IN_Y_BOT + 0.5`. research §B: BAT0004 "fits with **zero** length margin", 100 mm lead	Measure the cell (VE M7) and decide whic the leads exit. Add a channel (e.g. 3 × 1.5 groove) in the back or along a bay side, stowage spot for the on the PCB top besi (5.6 mm headroom Model the lead as a context body
R2-D19	NOTE	Silk legibility. KiCad footprint silk lines are 0.12 mm, and the design text is 0.8 mm high × 0.12–0.13 mm stroke. That is below or at PCBWay's commonly quoted legend minimums (~0.15 mm line, 0.8 mm height; UNVERIFIED against today's page). Black legend on white mask also prints softer. Polarity-critical silk: J2 "+/−" (0.9/0.15, fine), the RGB "1"/chamfer, the 2835 brackets	Gerber apertures: main F.Silk C0.12/C0.13/C0.15. LED F/B.Silk C0.12/C0.13/C0.15	Acceptable for v0.1. v0.2, use ≥ 0.15 mm and ≥ 1.0 mm text f polarity marks
R2-D20	NOTE	The zip also contains the drill-map gerbers (`*-drl_map.gbr`), an empty `led-board-NPTH.drl` and an empty `main-board-B_Paste.gbp`. PCBWay's viewer may list them as unknown layers. Harmless	`unzip -l` of both zips. The zip is byte-identical to `gerbers/` (md5 checked)	Drop the map files f the zip, or mention t in the order note
R2-D21	NOTE	The J2 plug "keep-out" (`KEEPOUTS`, x 9.6–15.8, y 50–56) is a F.Cu rule area with tracks and vias **allowed**: four GND stitching vias (11.975/50.975, 11.975/53.975, 14.975/50.975, 14.425/54.525) and the F.Cu GND pour sit under the plug. Vias are tented and the plug sits on mask, so it is harmless, but the comment overstates what it does	pcbnew: rule area F.Cu `noTracks False noVias False`. PTH drill hits	Leave as is. If R2-D1 enlarges the slot he those vias move any
R2-D22	NOTE	Round 26 mm board: PCBWay often routes small round boards with breakaway tabs. A nub on the edge eats the 0.2 mm radial pocket clearance (Ø26.4 pocket)	`moonlib.FIT_PCB = 0.2`	Ask for "no tab resid smooth edge" in the note, or sand the nu
R2-D23	NOTE	Handoff 01 §3 says "both are < 50 × 50 mm"; the main board is 35 × 62. The LED-board silk has name and revision but no date (pcb convention: name + rev + date), and both `.gbrjob` files carry Revision "rev?"	Handoff 01. `led-board-job.gbrjob`	Text fixes. Set the tit block rev/date so it reaches the job file drawings

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R2-D24	NOTE	Paste and mask details. 2835 paste apertures are 1:1 on the 2.2 × 2.2 mm cathode pad: heavy paste, so the LED may tilt or float. Mask webs are 0.25 mm (LED-board TSSOP-28/8) and 0.29 mm (USB-C CC/VBUS) at zero expansion, about 0.15 mm after a typical +0.05 mm fab expansion. That is fine for green, borderline for white mask	Gerber apertures. Mask-web script (`scripts/maskweb.py`)	Optionally reduce th 2835 paste aperture ~80 % or windowpa Ask PCBWay to keep mask dams betweer TSSOP pads
R2-D25	NOTE	LED-board J1 pad 1's outer corner is at r 11.84, beyond `moonlib.COMPONENT_R_MAX` 11.8 by 0.04 mm (copper only, no height). The silicone strain-relief blob (handoff 05 §3.4) could spread past r 12.0 onto the ledge and tilt the board	pcbnew radial scan: pad bbox max r 11.836 (J1.1). Ledge inner r 12.0	Handoff: keep the bl inside r 11 (toward t centre). Optionally r J1 0.3 mm inward in

Checked and found OK (with evidence)

Gerbers, drill and outline

- Both zips are byte-identical to `gerbers/`. Layer set: F/B Cu, Mask, Paste, Silk, Edge_Cuts (gm1), PTH/NPTH Excellon (METRIC, decimal, absolute), job file.
- Main board outline: closed, 35 × 62 mm, r 2 corners. The internal battery-lead slot is a separate closed loop on Edge_Cuts.
- LED board outline: a closed Ø26.000 circle made of two G02 arcs. Job-file thickness is 1.6 mm (main) and 1.0 mm (LED).
- NPTH holes: 2 × 0.9 mm (PCM12 pegs, which are Ø0.8) and 2 × 2.2 mm (M2). Board-file PTH: vias 0.3 mm, J4 1.0 mm, and USB slots.

Clearances and design rules

- Main board minimum track/space is 0.25/0.2 mm, vias are 0.6/0.3 mm (0.15 mm ring), and copper-to-slot is 0.29 mm.
- LED board: re-running DRC at 0.2 mm clearance finds a single spot at **0.186 mm** (> 6 mil). Minimum track is 0.2 mm.
- No via-in-pad on either board.
- Stored DRC/ERC reports show 0 errors, 0 unconnected and 0 parity issues. The only warnings are 2 silk_edge warnings on SW1.

LED board keep-out band. All copper (both pours, tracks, pads) is at r ≤ **11.999 mm**, and silk is at r ≤ 11.08 mm. So the clamp band r 12–13 is copper-free, the PETG ring (ID 24.0) bears only on mask, and the ledge (r 12.0–13.2) is clear of bottom parts (max r ~11.4 apart from the J1 pads). The pocket is Ø26.4.

Footprints against datasheets

Part	Footprint check	Polarity / pinout check
USB4125	CC 0.70 / VBUS 0.76 / GND 0.80 at 1.00 / 3.04 / 5.50, slots 0.6 × 1.2 in 1.1 × 1.7 pads at 8.64, body front on the board edge	n/a
PCM12	pins at 3.0 + 1.5 pitch, Ø0.9 holes for Ø0.8 pegs, 3.0 apart	knob 1.0 mm past the edge
PTS645SM43	9.5 / 6.4 × 5.8 / 3.2 layout matches exactly	rows 1–1 / 2–2 match C&K "1–2, 3–4"
JST SH	0.6 × 1.55 signal, 1.2 × 1.8 MP, 3.875 offset	opening toward the top edge
JST PH SM4-TB	n/a	opening toward +x; "+" silk at pin 1 (BAT_RAW)

Part	Footprint check	Polarity / pinout check
SOT-23 Q1–Q4	n/a	G = 1, S = 2, D = 3 per DMG2305UX; the nets give correct body-diode directions
MCP73831 SOT-23-5	n/a	1 STAT, 2 VSS, 3 VBAT, 4 VDD, 5 PROG
ATtiny3216-SN	SOIC300 (C04-00094), so SOIC-20W is correct	n/a
D1 SOD-123F, D3 SMA	n/a	pad 1 is the cathode, on VBUS and VSYS_RAW respectively
TSSOP-28	KiCad 1.475 × 0.4 at ±2.86 vs TI PW 1.5 × 0.45 at 5.8	n/a
TSSOP-8	footprint is for SOT505-1	n/a
Bridgelux 2835 (DS312 p.14)	pads 2.2 / 1.25 vs Bridgelux 2.07 / 1.17	bottom view "−" = the 1.97 mm terminal, i.e. **the large pad is the cathode**, which is KiCad pad 1 = K_Dn
Würth 150141M173100	land pattern 1.5 × 0.9 at 3.1 × 1.4 matches exactly	pad 1 = VLED
Custom 4-pad cable footprint	1.3 × 2.4 at 2.0 pitch, no paste	G/V/D/C silk matches pads 1–4 (GND, VLED, SDA, SCL), and matches the Qwiic colours in handoff 05

BOM and CPL

- Main board: 36 placed parts, with BOM and CPL agreeing one-to-one.
- LED board: 17 BOM parts plus J1 (R2-D9). Bottom parts are flagged Bottom; DNP and test points are excluded.
- MPNs match `hardware/parts.py`.

Enclosure fit (by construction)

- The main-board mounting holes coincide with the `pack_front` bosses, because both are derived from `MOUNT_HOLES`.
- The Ø5 boss footprints are clear of all parts. The nearest part is the SW4 cap flange, 0.2 mm away (R2-D11).
- The button, charge-LED, USB-C and slide-switch openings are derived from `PLACE`, so they track the board.
- J3 is centred on the neck axis (X 0). The plug's cable reaches the 6.8 mm bore at Z ≈ 12.2, inside the bore (Z 5.9–12.7).
- Other top-side heights against TOP_CLEAR 5.6 mm: PTS645 4.3 (caps sized to it), USB4125 ~3.2, SOIC-20W 2.65, SMA 2.4, SH 2.95, PCM12 1.4.

Still UNVERIFIED (needs parts in hand or the live PCBWay page)

- Gooseneck thread length (R2-D3).
- PCBWay's current figures for legend line/height, plated-slot and annular-ring minimums.
- Real pouch dimensions and lead exit (R2-D18).
- USB-C overmold shapes of Matt's cables (R2-D14).
- Exact PCM12 actuator Z band (the window spans −0.4 to +2.2 mm about the PCB top, so it is very likely fine).